

AgriSmart AI

Bonus Module A — Crop Recommendation Model

SIH 2026 (Internal Hackathon) · L.J. Institute of Engineering and Technology [C-433]

TASK	Multi-class crop recommendation — given 7 soil/climate readings, recommend the single best-suited crop out of 22 classes (rice, maize, jute, cotton, coconut, papaya, orange, apple, muskmelon, watermelon, grapes, mango, banana, pomegranate, lentil, blackgram, mungbean, mothbeans, pigeonpeas, kidneybeans, chickpea, coffee).															
DATASET & SPLIT	Source: Crop_recommendation.csv (Kaggle — atharvaingle/crop-recommendation-dataset). 2,200 rows × 7 numeric features (N, P, K, temperature, humidity, ph, rainfall) + label. No missing values, no duplicate rows; classes are perfectly balanced at 100 samples each. Split: scikit-learn train_test_split, test_size = 0.2, random_state = 42 → 1,760 train / 440 test (held-out, untouched during fitting). Preprocessing: MinMaxScaler fit on train only, followed by StandardScaler fit on the min-max-scaled train set; both transformers applied to test with transform() only (no re-fitting) in the final pipeline.															
MODEL / APPROACH	Ten classifiers were benchmarked with default hyperparameters on the same train/test split to select a candidate: Logistic Regression, Gaussian Naive Bayes, SVC, KNN, Decision Tree, Extra Tree, Random Forest, Bagging, Gradient Boosting, and AdaBoost. Final model: RandomForestClassifier(n_estimators=100, random_state=42, n_jobs=-1), trained on the scaled 1,760-row training set. Inference exposes predict() for the top class plus predict_proba() to rank the top-3 candidate crops with confidence scores. Model and both scalers are persisted (joblib: crop_model.pkl, crop_minmax_scaler.pkl, crop_standard_scaler.pkl) for reuse at inference time.															
METRIC & RESULT	<table><thead><tr><th>Metric (test, n=440)</th><th>Value</th><th>Notes</th></tr></thead><tbody><tr><td>Accuracy</td><td>99.32% (437/440)</td><td>3 misclassifications</td></tr><tr><td>Macro-F1</td><td>0.9926</td><td>22 balanced classes</td></tr><tr><td>Macro-Precision</td><td>0.9926</td><td>—</td></tr><tr><td>Macro-Recall</td><td>0.9933</td><td>—</td></tr></tbody></table> Weakest classes: rice (recall 0.89 — 2 of 19 confused with a similar-climate crop) and jute / lentil (precision ~0.92, driven by 1 borderline sample each); all other 19 classes score 1.00 across precision, recall and F1.	Metric (test, n=440)	Value	Notes	Accuracy	99.32% (437/440)	3 misclassifications	Macro-F1	0.9926	22 balanced classes	Macro-Precision	0.9926	—	Macro-Recall	0.9933	—
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BASELINE	No organizer-issued baseline applies to this bonus module (it is only defined for the core disease-detection task). As an internal baseline, the same split was scored with 9 other default-hyperparameter classifiers: • Gaussian Naive Bayes: 99.55% — narrowly the single best accuracy of all models tried. • Random Forest (selected): 99.32% • Bagging: 98.86% · Decision Tree: 98.64% · Gradient Boosting: 98.18% • SVC: 96.82% · Logistic Regression: 96.36% · KNN: 95.91% · Extra Tree: 90.91% · AdaBoost: 14.55% (default estimator unsuited to this multi-class task) Random Forest was carried forward for its balanced precision/recall across all 22 classes and native predict_proba() support for top-3 ranked recommendations — Gaussian NB's marginal edge is a reasonable open question for further tuning.															
LIMITATIONS	• Dataset is a clean, synthetic-feeling benchmark (perfectly balanced, no missing/noisy values) — every model tried except AdaBoost exceeded 95% accuracy, so this does not yet demonstrate generalisation to noisier real farm-sensor or manual-entry data. • No cross-validation or hyperparameter search was run for the final Random Forest — default n_estimators=100 only; a k-fold CV and a small grid/random search would give a more robust estimate than a single 80/20 split. • An early inference draft in the notebook re-fit the scalers with .fit_transform() on each single incoming sample (data leakage / undefined behaviour on n=1 input); this was identified and corrected in the final pipeline, which loads scalers fit once on the training set and only calls .transform() at inference. • Feature set omits location, season, and previous-crop context called for in the PS's Bonus-A description.															

Note: covers Bonus Module A (Crop Recommendation) only. The core, primary-scored metric is macro-F1 on the crop-disease image classifier, judged separately on the organizers' held-out field set.